# YANGYANG LI

University of Science and Technology of China (USTC) +86-15797758378
96 Jinzhai Rd., Hefei, Anhui 230026, China liyangyang.xyz@gmail.com

## Educational Background

**USTC**, BS in Department of Modern Physics, School of Physical Sciences 09/2021 - 07/2025
Major: Theoretical Physics, GPA: 4.08/4.3, Rank: 4/150, 1/48

## Experiences

### C-R-T Fractionalization in the First Quantized Hamiltonian Theory 07/2024 - 12/2024

Undergraduate Researcher,

**Supervisors:** Yi-Zhuang You, Juven Wang, **Collaborators:** Zheyan Wan, Shing-Tung Yau

· **Our goal: Examine free fermion's C-R-T and internal symmetry group and find the relations between different dimensional cases.**

· Redefined the Majorana fermion as the representation of Clifford algebra to discuss the CRT-internal symmetry of free Majorana fermion in all dimensions (especially $d+1 = 5, 6, 7 \mod 8$).

· **Assigned CRT and internal symmetries** for Majorana and Dirac fermion, **examine their group structure**.

· Assigned all possible fermion bilinear mass terms for Majorana and Dirac fermion, and **examined the interplay of the mass manifold (spanned by different masses) and the symmetries**.

· **Demonstrated the relations of CRT-internal symmetry groups in different dimensions** using the domain-wall reduction method.

· Our research results (arXiv:2412.11958) have appeared on arXiv.

### Many-Body Lattice C-R-T Symmetry: Fractionalization, Anomaly, and Symmetric Mass Generation 07/2024 - 12/2024

Undergraduate Researcher, **Supervisors:** Yi-Zhuang You, Juven Wang

· **Our goal: Classify symmetric mass generation (gapping many-body interaction with full CPT-internal symmetries) on lattice for generalized dimensions and find the explicit gapping interaction.**

· Derived the CPT-internal symmetry group for lattice Majorana and Dirac fermion, (with staggered fermion Hamiltonian) and found their correspondence in free model. The IR and UV symmetry matching includes emergent symmetries and emanant symmetries.

· **Successfully derived the classification for 0+1d, 1+1d, 2+1d, and 3+1d symmetric mass generation** by finding anomalies in many-body invariant group, searching for explicit interaction terms (using Mathematica codes) that can gap the system, **and generalizing it to higher dimensions**.

· **Numerically checked the gapping interaction terms** using Jordan-Wigner transformation in many-body Hilbert space.

· **Analytically checked the gapping interaction terms** using stabilizer analysis.

· Our research results (arXiv:2412.19691) have appeared on arXiv.

### Floating Edge Bands in the BHZ model with Altermagnetism 06/2024 - 08/2024

Undergraduate Researcher, **Supervisor:** Song-Bo Zhang

· **Our goal: Give a thorough analytical result of floating edge band in a simplified model.**

· Discovered floating edge bands (completely detached from bulk bands throughout Brillouin zone) in BHZ model with altermagnetism.

· **Derived the phase diagram analytically** using Chern number and winding number at high symmetry points.

· Derived the spin-resolved density of state for different phases using iterative numerical methods of surface Green's function.

· **Analytically calculated the non-perturbative band structure and wave function** for floating edge bands.

· Verified the robustness of floating edge band state numerically against geometric rotation, energy asymmetry, and spin coupling.

· Our research results (PhysRevB.111.045106) have been **published on Phys. Rev. B**.

### Publications

[1] **Y.-Y. Li** and S.-B. Zhang, Floating edge bands in the Bernevig-Hughes-Zhang model with altermagnetism, Phys. Rev. B 111, 045106 (2025).

[2] **Y.-Y. Li**, Z. Wan, J. Wang, S.-T. Yau, and Y.-Z. You, C-R-T fractionalization in the first quantized Hamiltonian theory, arXiv e-prints , arXiv:2412.11958 (2024)[cond-mat.str-el].

[3] **Y.-Y. Li**, J. Wang, and Y.-Z. You, Quantum many-body lattice C-R-T symmetry: fractionalization, anomaly, and symmetric mass generation, arXiv e-prints , arXiv:2412.19691 (2024)[cond-mat.str-el].

## Skills

**Knowledge:** Condensed matter field theory, Topological insulator, Clifford algebra and lattice fermion theory, Iterative surface Green's Function method, Anomaly analysis, Differential geometry.

**English:** TOEFL: 103 (29 R + 28 L + 23 S + 23 W), GRE 160 (Verbal) + 170 (Quantitative) + 3.5 (Writing)

## Awards

· 862 Alumni Excellent Freshman Scholarship ( 3/379 ) 11/2022
· Excellent Student Scholarship Silver Award ( 10% ) 12/2023
· National Scholarship ( 11/150 ) 11/2024
· Guo Moruo Scholarship ( 35/1950 ) 03/2025